HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Monitored **AMHS hardware performance** and reliability using data analytics, identifying trends that led to a **15% improvement** in system availability.
- Provided engineering support as the **hardware subject matter expert** to technicians and manufacturing teams, resolving complex automation system issues.
- Applied advanced **troubleshooting techniques** to diagnose and resolve automation system problems, enhancing operational efficiency and throughput.
- Collaborated with technicians and cross-functional teams to implement **predictive and preventative maintenance programs** for manufacturing equipment.
- Managed project efforts for system upgrades, including **planning, budgeting, and forecasting** for equipment installations and factory ramp activities.
- Created improvement roadmaps by analyzing **system performance metrics**, focusing on enhancing equipment availability and productivity.
- Containerized Java Spring Boot applications using **Docker** and deployed them to **AWS EC2 instances**, improving deployment consistency.

SKILLS

Domain Expertise – Automated Material Handling System (AMHS), Semiconductor Manufacturing, Manufacturing Equipment, Factory Operations, Production Systems

Engineering & Analysis – Data Analytics, Statistical Process Analysis, Mechanical Engineering, Electrical Engineering, Failure Analysis, Problem-Solving, Reliability Engineering, Predictive Maintenance, Preventative Maintenance, Continuous Improvement

Systems & Tools – Automation Systems, PLC, Control Systems, AMHS Hardware, Manufacturing Technology, JIRA, Confluence

Programming & Scripting – Python, Java, SQL, Spring Boot, REST APIs, Shell Scripting

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, CI/CD, Jenkins, Git

Project Management – Project Management, Budgeting, Forecasting, Stakeholder Management, Vendor Management, Change Control

PROJECTS

AMHS Reliability Enhancement Project (HCL Technologies)

- Analyzed historical performance data from **Splunk and Dynatrace** to identify root causes of AMHS downtime, leading to targeted improvements.
- Developed Python scripts to automate the collection and initial analysis of sensor data, reducing manual data review time by **40%**.
- Implemented enhanced monitoring dashboards in **Dynatrace** to provide real-time visibility into critical AMHS hardware components.

Automated Data Pipeline for Manufacturing Analytics (HCL Technologies)

- Engineered an automated data pipeline using **Python and PostgreSQL** to ingest and transform manufacturing operational data for analysis.
- Designed and implemented ETL processes that consolidated data from multiple sources, improving data accuracy for reporting by **~20%**.
- Created scheduled reports and alerts using **SQL queries** on PostgreSQL, enabling faster response to production anomalies.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)